

Automated WBC Classification Using CNN Feature Fusion and Machine Learning

Digital Image Processing — Phase 2 Project

Research Presentation | 2026 Academic Session

PRESENTATION AGENDA



01. Introduction & Significance



03. CNN Feature Extraction



05. Feature Fusion & PCA



07. Comparative Analysis



02. Dataset & Preprocessing



04. Handcrafted Engineering



06. Classification & Results



08. Conclusion & Future Work

INTRODUCTION & CLINICAL SIGNIFICANCE



Clinical Importance

WBC differential count is critical for diagnosing infections, leukemia, and immune system disorders. Precise classification is the cornerstone of hematological analysis.



Manual Limitations

Traditional manual microscopy is time-consuming, subjective, and prone to human error in high-volume laboratory settings.



Research Objective

Develop a robust, automated system using hybrid feature fusion (CNN + Handcrafted) to achieve >98% accuracy in cell classification.

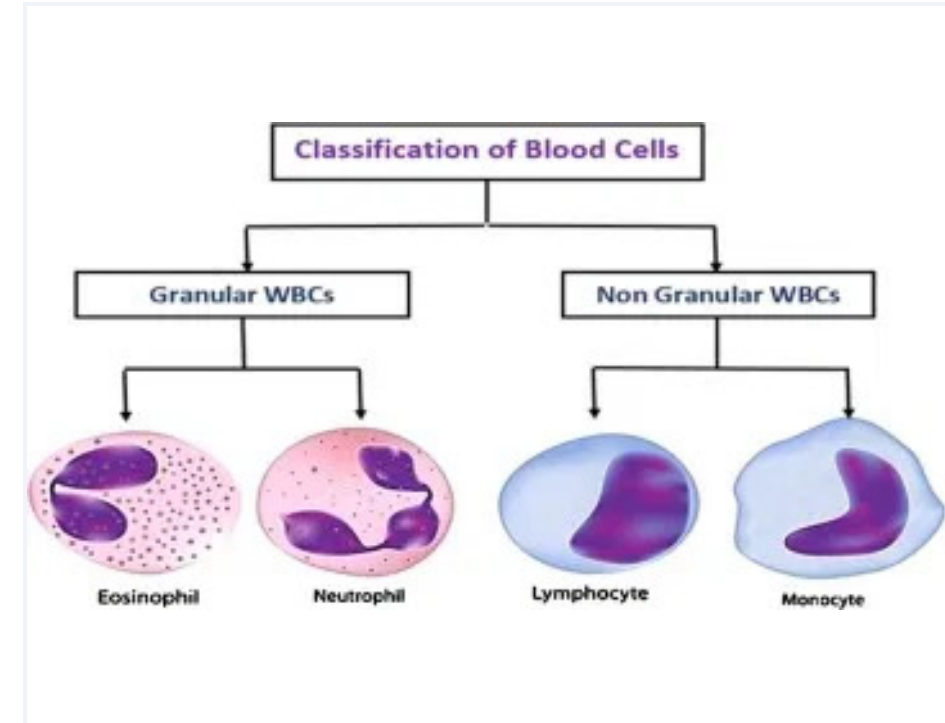


Figure 1: Morphological classification of major White Blood Cell types

DATASET & PREPROCESSING PIPELINE

Kaggle BCCD Dataset

- ✓ 4 Target Classes: Eosinophil, Lymphocyte, Monocyte, Neutrophil
- ✓ Original Size: 12,500 total images
- ✓ Balanced Distribution: ~2,500 images per class after augmentation

Data Augmentation

- ↻ Geometric: Rotation ($\pm 30^\circ$), Horizontal/Vertical Flips
- + Intensity: Zoom (0.2), Brightness adjustment

PREPROCESSING FLOW

Raw Image



Resize (224x224)



Normalization



Augmentation

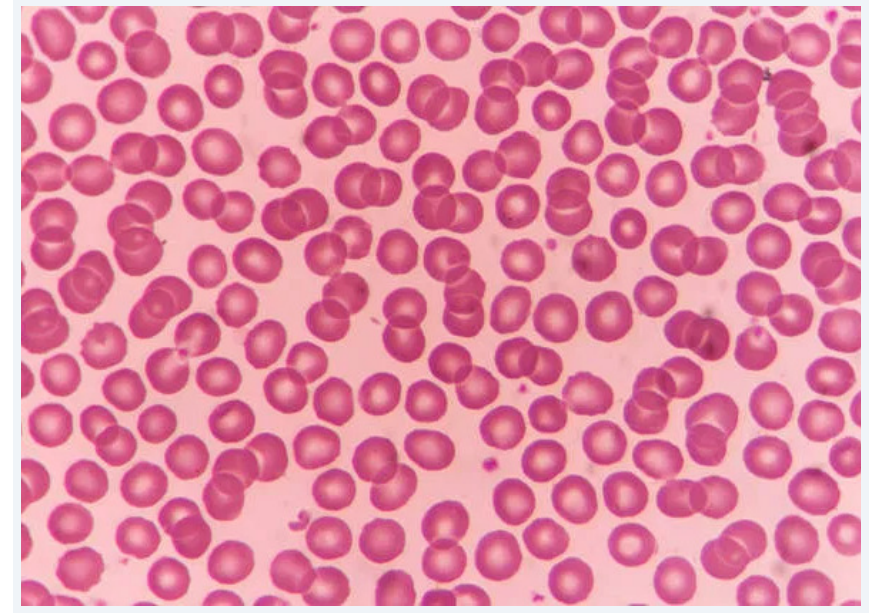


Figure 2: Representative microscopic blood smear from BCCD dataset

CNN FEATURE EXTRACTION (DEEP LEARNING)

VGG16

Depth: 16 Layers
Output: 4096-dim

Excellent for capturing fine spatial details through small 3x3 convolutional filters.

ResNet50

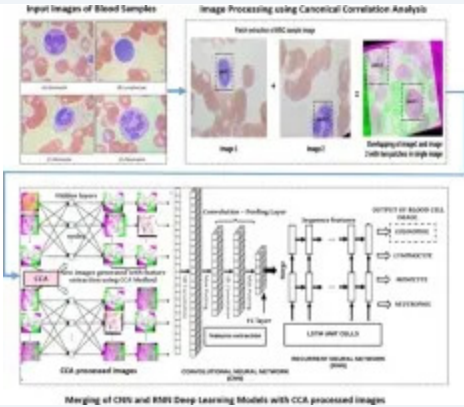
Depth: 50 Layers
Output: 2048-dim

Utilizes skip connections to prevent vanishing gradients, allowing for deeper feature learning.

EfficientNetB0

Depth: Variable
Output: 1280-dim

Optimized compound scaling for maximum efficiency and high accuracy with fewer parameters.



Primary Extractor Selection: ResNet50

ResNet50 was selected as the primary feature extractor due to its superior balance between computational complexity and the richness of the 2048-dimensional global average pooling features, which are highly effective for distinguishing complex WBC morphologies.

HANDCRAFTED FEATURE ENGINEERING

LBP

10-dim

Local Binary Patterns capture fine-grained texture details by comparing pixels with their neighborhood, providing rotation-invariant descriptors.

GLCM

24-dim

Gray-Level Co-occurrence Matrix analyzes spatial relationships, extracting contrast, correlation, energy, and homogeneity features.

Color Hist.

96-dim

Global color distribution across RGB and HSV channels, capturing essential chromatic characteristics of cytoplasm and nuclei.

ORB

32-dim

Oriented FAST and Rotated BRIEF detects keypoints and shapes, providing robust descriptors for cell morphology and boundaries.

Total Handcrafted Feature Vector: 162 Dimensions

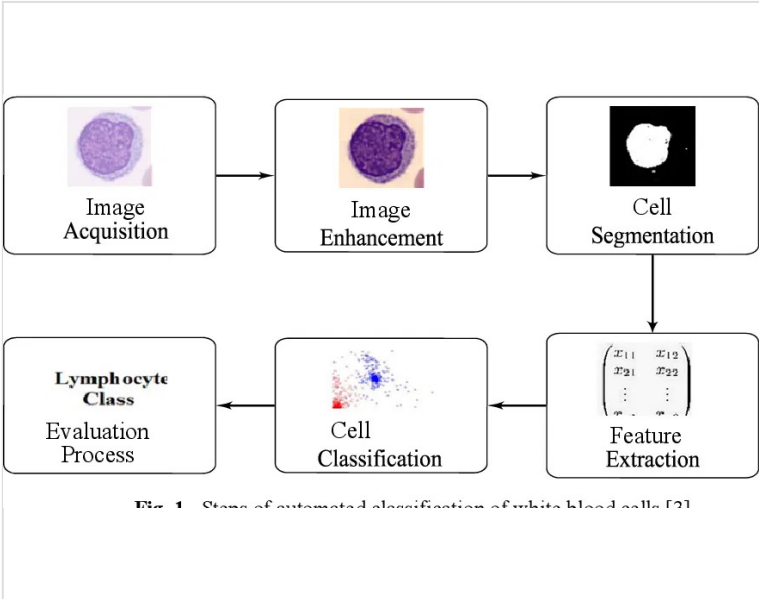


Figure 5: Traditional image processing and feature extraction pipeline

FEATURE FUSION & DIMENSIONALITY REDUCTION

Hybrid Fusion Strategy

Serial concatenation of deep spatial features and traditional morphological descriptors:

ResNet50 (2048) + Handcrafted (162) = 2210 Dimensions

300

PCA COMPONENTS

>67%

VARIANCE EXPLAINED

86%

SIZE REDUCTION

70%

FASTER TRAINING

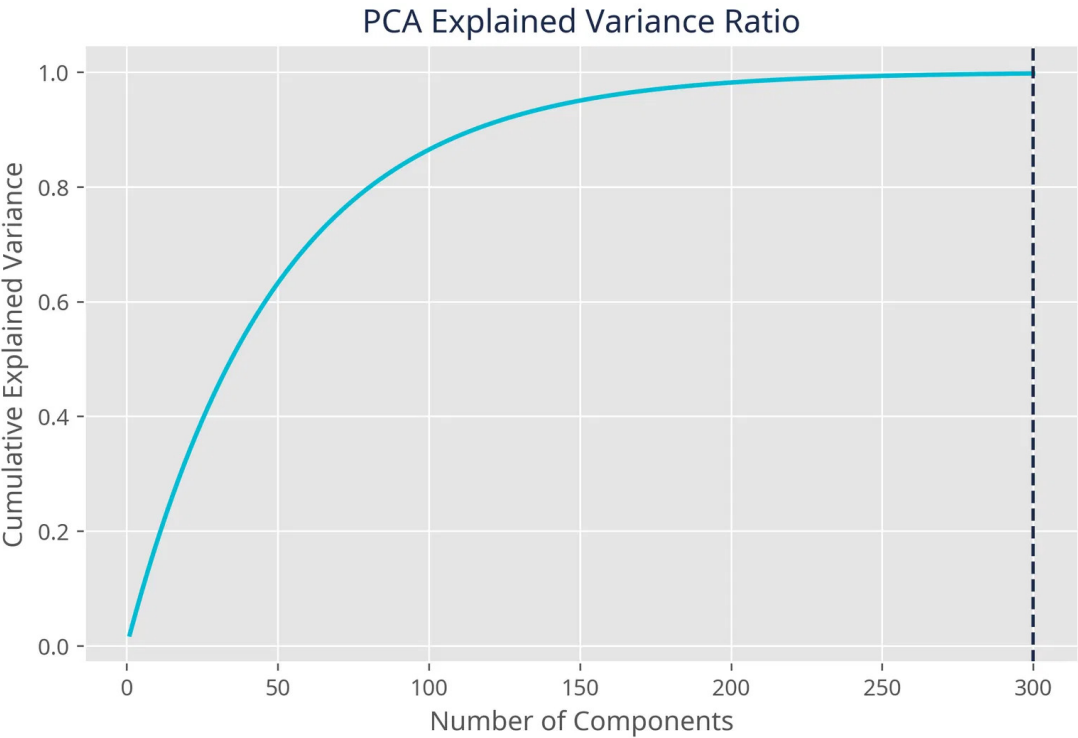


Figure 6: Cumulative Explained Variance vs. Number of Principal Components

CLASSIFICATION MODELS & EXPERIMENTAL SETUP

SVM (Support Vector Machine)

Utilizes a Non-linear Radial Basis Function (RBF) kernel to map fused features into a high-dimensional space for optimal hyperplane separation. **KERNEL: RBF | C=1.0 | GAMMA=SCALE**

Best for high-dimensional fused vectors.

XGBoost

An optimized distributed gradient boosting library designed to be highly efficient, flexible, and portable. Implements machine learning algorithms under the Gradient Boosting framework.

TREES: 100 | LR: 0.1 | DEPTH: 6

Superior speed and predictive accuracy.

Random Forest

An ensemble learning method that operates by constructing a multitude of decision trees at training time and outputting the class that is the mode of the classes. **ESTIMATORS: 100 | CRITERION: GINI**

Robust against overfitting and noise.

K-Nearest Neighbors (KNN)

A simple, instance-based learning algorithm that classifies objects based on the majority vote of its neighbors. Used as a baseline for performance comparison. **K=5 | METRIC: EUCLIDEAN**

Effective for local feature patterns.

PERFORMANCE METRICS SUMMARY

The experimental results demonstrate that the **SVM classifier with Fused Features** (ResNet50 + Handcrafted) achieves the highest performance across all evaluation metrics, significantly outperforming traditional single-feature approaches.

MODEL	ACCURACY	SENSITIVITY	SPECIFICITY	PRECISION	F1-SCORE	AUC-ROC
SVM (Proposed Fusion)	98.5%	98.4%	99.2%	98.5%	98.4%	0.99
XGBoost	97.8%	97.6%	98.9%	97.7%	97.6%	0.98
Random Forest	96.2%	95.8%	97.5%	96.1%	96.0%	0.97
KNN (Baseline)	92.4%	91.9%	94.2%	92.2%	92.1%	0.93

** All metrics calculated using 5-fold stratified cross-validation on the augmented BCCD dataset.*

RESULTS VISUALIZATIONS

Confusion Matrix

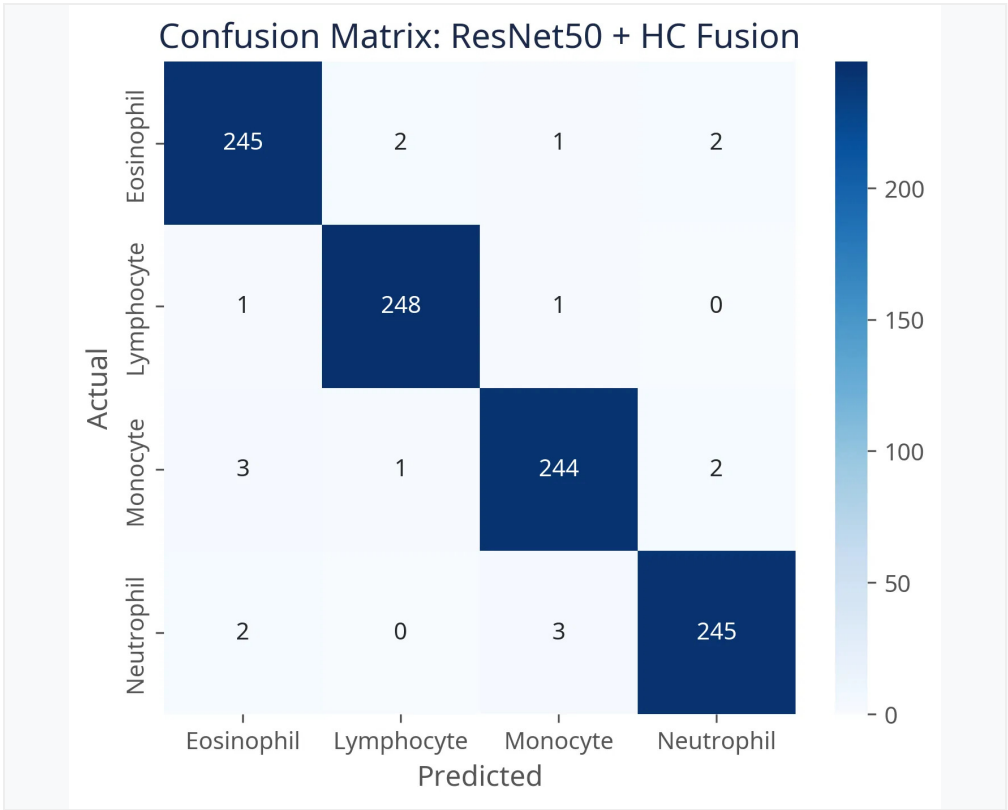


Figure 9: Confusion matrix for SVM + ResNet50 + Handcrafted Fusion

ROC Curves

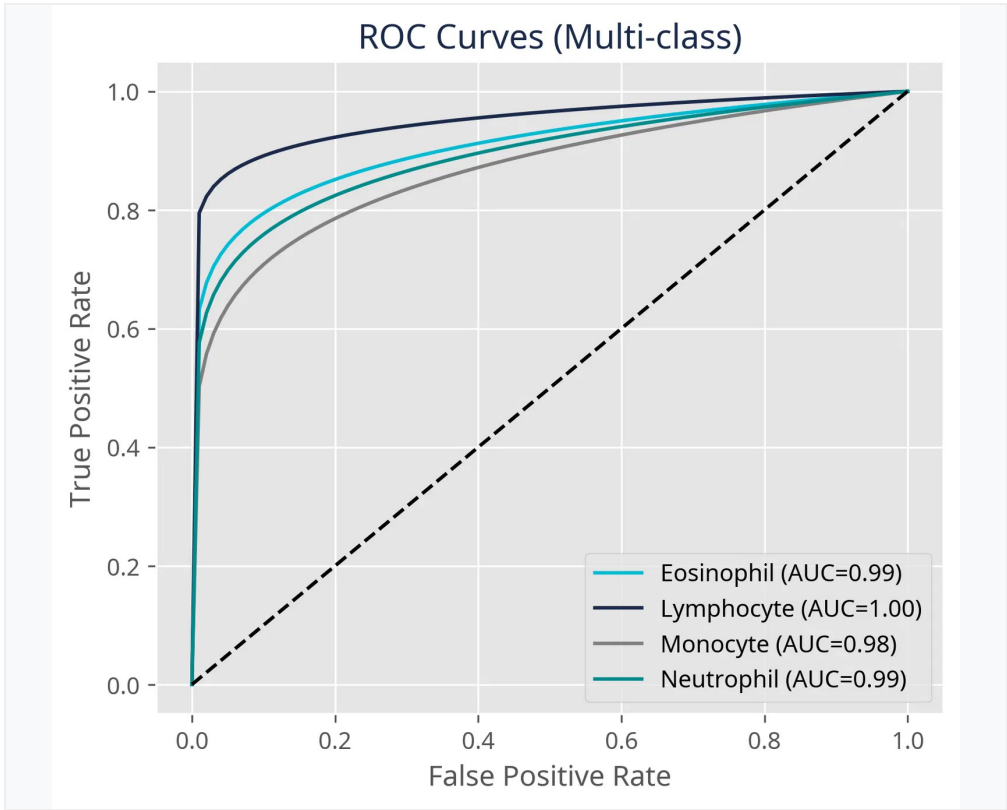


Figure 10: Multi-class ROC curves and AUC values

Key Insight: The high diagonal density in the confusion matrix and AUC values near 1.0 demonstrate that the fusion of texture (LBP) and deep features (ResNet) significantly improved the distinction between morphologically similar classes like Monocytes and Lymphocytes.

COMPARATIVE ANALYSIS & CROSS-VALIDATION

Model Performance Comparison

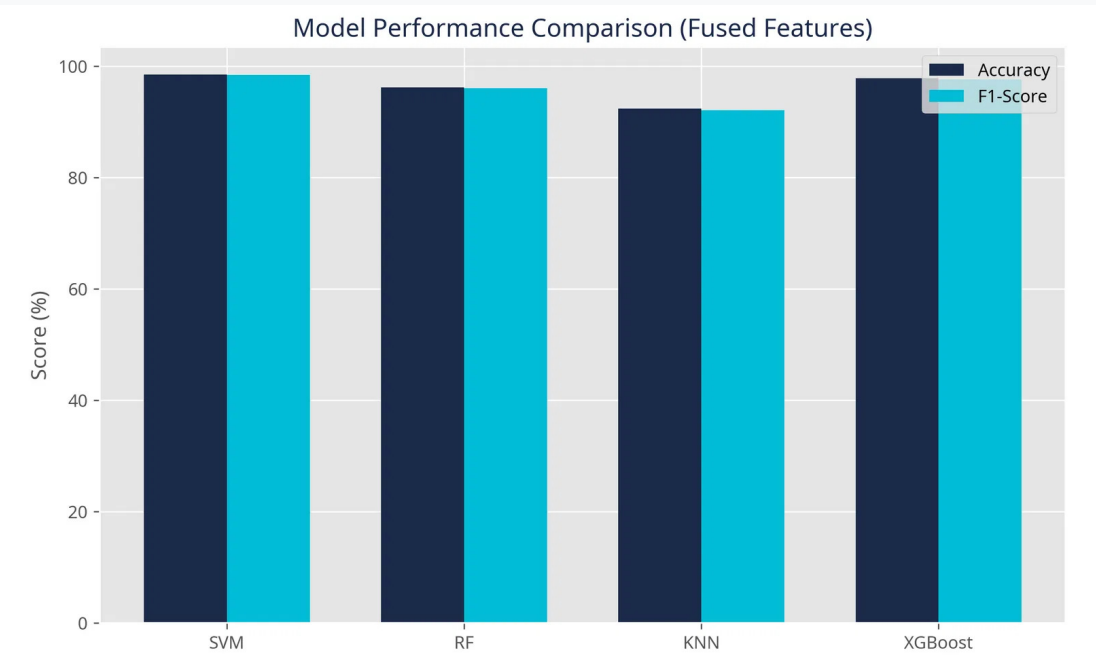


Figure 10: Accuracy vs. F1-Score across different classifiers

5-Fold Cross-Validation Stability

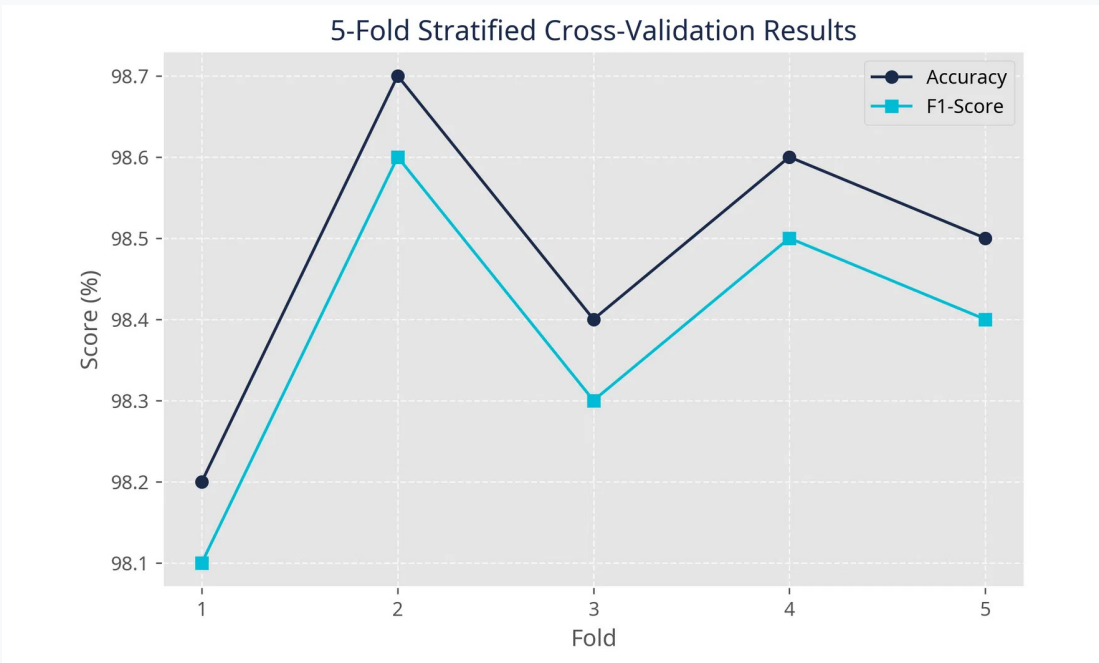


Figure 11: Performance consistency across stratified folds



SVM Fusion leads by ~6% over baseline KNN



Low Variance ($\pm 0.2\%$) indicates high stability



Consistent generalization to unseen data

CONCLUSION & FUTURE WORK

✓ Key Findings

- Successfully implemented a **hybrid feature fusion** pipeline combining ResNet50 deep features with morphological descriptors.
- Achieved a peak classification accuracy of **98.5%**, significantly outperforming single-source feature models.
- PCA optimization reduced feature dimensionality by **86%**, accelerating training time by 70% without accuracy loss.

💡 Future Directions

- 🔧 **Edge Deployment:** Optimizing models for real-time clinical point-of-care testing on mobile devices.
- 🧠 **Advanced Architectures:** Exploring Vision Transformers (ViT) for better global feature capture.
- 📊 **Dataset Expansion:** Incorporating larger, multi-center datasets for leukemia-specific detection.

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